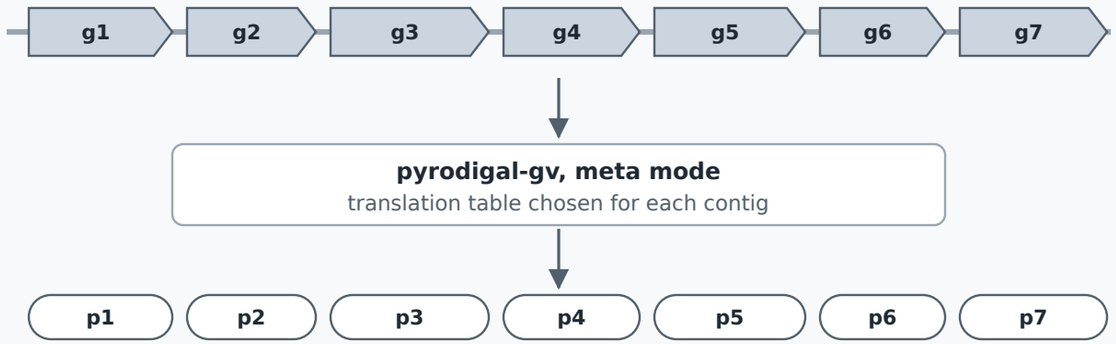
# phu avger: finding auxiliary viral gene (AVG) candidates in viral contigs

## A Gene prediction

Trusted viral contig (FASTA)



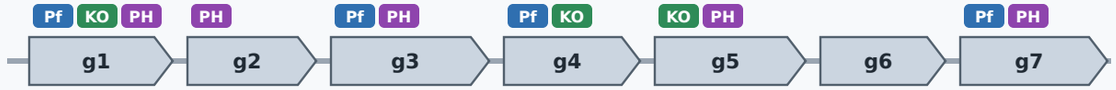

Translated proteins (at least 30 aa by default)

*Predictions are cached and shared with phu screen and phu jack*

## B Two HMM annotation tracks (PyHMMER)

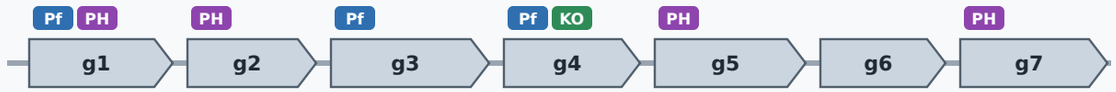
Pfam · KOfam · PHROG

**Relaxed track** E-value ≤ 1e-5; best hit per protein in each database




→ feeds V / VL scoring (panel C)

**Strict track** Pfam GA cutoffs · KOfam KO thresholds · PHROG bit score ≥ 60, coverage ≥ 0.4



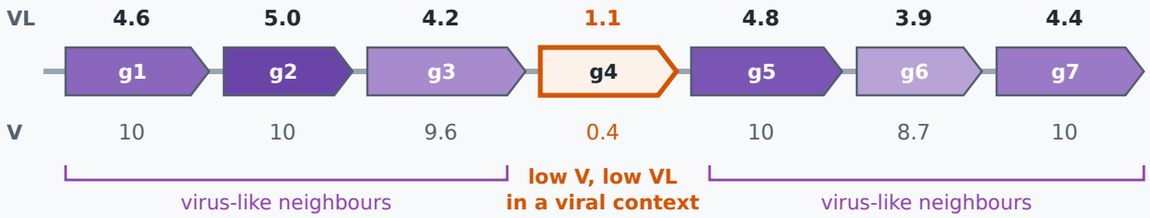

→ feeds functional calls and reference matching (panel D)

*Each database is searched once; the two tracks differ only in the thresholds applied to the hits*

*Pfam, KOfam and PHROG are scored and averaged independently; values are never mixed*

## C Viral-likeness scores (V-Score-Search tables)

Relaxed-track hits are looked up in the V / VL table, separately for each database




**contig AVL (per database) = mean VL of the contig's scored genes**

here: (4.6 + 5.0 + 4.2 + 1.1 + 4.8 + 3.9 + 4.4) / 7 = 4.0

**V-score** how exclusively a protein family occurs in viruses (10 = virus only)

**VL-score** V-score weighted by how often the family is found in viruses

**AVL** average VL across a contig, per database: the viral context of a gene

VL shading: lower … higher  below the gene cutoff

## D Candidate test and curation

**Candidate test, per database**

| | | |
|---|---|---|
| contig AVL | **> 3** | Pfam, KOfam and PHROG are each tested on their own |
| gene VL | **< 3** | |
| gene V | **< 10** | |

fails in all → **not_candidate**

passes in at least one database → **AVG candidate**

1. **Match strict hits to the CheckAMG positive lists**
   AMG: Pfam, KOfam · APG and AReG: Pfam, KOfam, PHROG
2. **AMG weight ≥ 0.6 (--min-amg-weight)**
   applies to AMG evidence only
3. **Class filters (--filter-mode standard | strict | none)**
   an essential or structural hit in any database, PHROG included, removes the class

**Final status**

classified: putative_amg · putative_apg · putative_areg

class_conflict · filtered · below_amg_weight
unclassified_candidate · not_candidate

## Outputs

| gene-annotations.tsv | avg-candidates.tsv | avg-predictions.tsv | avg-evidence.tsv | .phu/run.json |
|---|---|---|---|---|
| every predicted gene | genes passing the test | one final named class | hits and reference matches | parameters and database versions |

*All calls are computational candidates for prioritisation, not confirmed auxiliary genes.*

Scores: Zhou et al. 2026 · reference tables: CheckAMG · terminology: Martin et al. 2025 · gene values shown are illustrative